

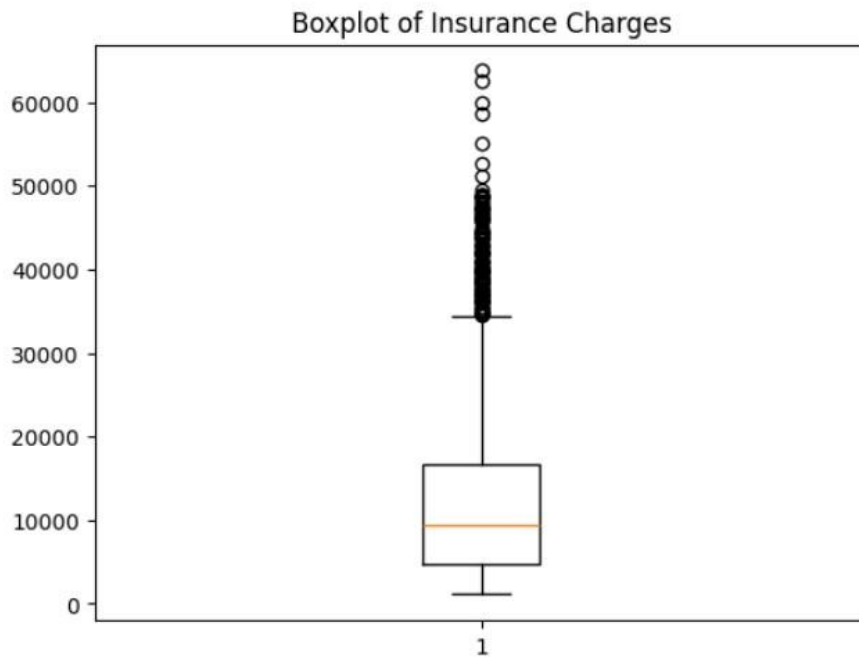
Missing value analysis , fixing missing values and outliers in dataset

Program:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
df=pd.read_csv('/content/archive (7).zip')
df.head(), df.info()
total_missing= df.isnull().sum().sum()
missing_per_column = df.isnull().sum()
summary = df.describe(include='all')
total_missing, missing_per_column, summary
df_mean = df.copy()
df_mean['bmi'] = df_mean['bmi'].fillna(df_mean['bmi'].mean())
df_median = df.copy()
df_median[' region'] = df_median[' region'].fillna(df_median['
region'].median())
df_mode = df.copy() df_mode['sex'] =
df_mode['sex'].fillna(df_mode['sex'].mode()[0])
df_ffill = df.fillna(method='ffill')
df_bfill = df.fillna(method='bfill')
numeric_cols = ['age', 'bmi', 'children', 'charges']
plt.figure()
plt.boxplot(df['charges'])
plt.title("Boxplot of Insurance Charges") plt.show()
```

OUTPUT:

...



```
Q1 = df[numeric_cols].quantile(0.25)
Q3 = df[numeric_cols].quantile(0.75)
IQR = Q3 - Q1
outliers_iqr = ((df[numeric_cols] < (Q1 - 1.5 * IQR)) | (df[numeric_cols] > (Q3 + 1.5 * IQR))).sum()
```

```
z_scores = np.abs(stats.zscore(df[numeric_cols]))
outliers_z = (z_scores > 3).sum(axis=0)
```

```
lower = df[numeric_cols].quantile(0.01)
upper = df[numeric_cols].quantile(0.99)
outliers_percentile = ((df[numeric_cols] < lower) | (df[numeric_cols] > upper)).sum()
```

outliers_iqr, outliers_z, outliers_percentile

```
(age      0
 bmi      9
 children 0
 charges  139
 dtype: int64,
 array([ 0,  4, 18,  7]),
 age      0
 bmi      28
 children 0
 charges  28
 dtype: int64)
```